Measurement of the Level of Poverty

Contemporary Economic Issues: ECON 204

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Introduction

- Poverty measurement is essential for assessing economic deprivation and informing policy decisions.
- Two broad approaches:
    1. **Absolute Poverty**: Defined as a fixed threshold, such as living on less than $2.15 per day (PPP-adjusted). This approach considers the minimum income required for basic human needs.
    2. **Relative Poverty**: Defined in relation to the overall income distribution of a society. Individuals are considered poor if they earn significantly less than the median income.

Absolute Poverty Measures

Definition: Absolute poverty measures assess whether individuals or households fall below a fixed income threshold, regardless of societal income levels.

Example: Poverty Line

Suppose a country sets a poverty line at $1,000 per year per person. Individuals earning below this threshold are considered poor.

Headcount Ratio (HCR)

Definition: The Headcount Ratio (HCR) measures the proportion of the population living below the poverty line.

Formula:

$$HCR = \frac{q}{N} \tag{1}$$

where:

- q = number of people below the poverty line
- N = total population

Example Calculation:

- Population: 10,000
- Poor individuals: 2,500
- $HCR = \frac{2{,}500}{10{,}000} = 0.25$ or 25%

Poverty Gap Index (PGI)

Definition: The Poverty Gap Index measures the intensity of poverty by calculating how far below the poverty line poor individuals are, on average.
Formula:

$$PGI = \frac{1}{N} \sum_{i=1}^{q} \frac{(Z - Y_i)}{Z} \tag{2}$$

where:

- Z = poverty line
- Y_i = income of individual i

Example:

- $Z = 1,000, \; Y_A = 800, \; Y_B = 700, \; Y_C = 500$
- $PGI = \frac{(0.20+0.30+0.50)}{3} = 0.333$ or 33.3%

Multidimensional Poverty Index (MPI)

Definition: The MPI assesses poverty using multiple indicators, including education, health, and living standards. A person is considered **multidimensionally poor** if they are deprived in multiple dimensions simultaneously. Each indicator represents an essential aspect of well-being, and being deprived in multiple indicators suggests a deeper level of poverty.
Example Data:

Person	Health	Education	Living Standards
A	1	0	1
B	0	1	0
C	1	1	1

Multidimensional Poverty Index (MPI)

Interpretation:

- A value of 1 in an indicator means the person is deprived in that dimension.
- A value of 0 means they are not deprived.
- The **MPI Poverty Threshold** is typically set at 0.33, meaning a person must be deprived in at least one-third of the total indicators to be classified as multidimensionally poor.
- In this example, **Person A and C** are multidimensionally poor since they experience deprivation in multiple dimensions.

Policy Recommendations

- Implement targeted cash transfer programs to alleviate extreme poverty.
- Improve access to education and healthcare to enhance human capital.
- Support microfinance initiatives for self-employment opportunities.
- Strengthen labor market policies to ensure fair wages and employment security.

Conclusion

- Measuring poverty requires both absolute and relative approaches.
- Combining income-based and multidimensional measures provides a fuller picture.
- Effective policies require a deep understanding of poverty severity and distribution.

Problem 1: Poverty Line Calculation

Question: The poverty line in Country X is set at \$1,200 per month for a household of four people. The average income of a family in this country is \$1,800 per month. Calculate the poverty gap ratio for the following families:

1. Family A: Monthly income = \$1,100
2. Family B: Monthly income = \$1,400

Problem 1: Poverty Line Calculation

Solution: The poverty gap ratio for a family is calculated as:

$$\text{Poverty Gap Ratio} = \frac{\text{Poverty Line} - \text{Income of the Household}}{\text{Poverty Line}}$$

Family A: Income = \$1,100, Poverty Line = \$1,200 Poverty Gap = \$1,200 - \$1,100 = \$100 Poverty Gap Ratio = $\frac{100}{1200} = 0.0833$
Family B: Income = \$1,400, Poverty Line = \$1,200 Poverty Gap = \$1,200 - \$1,400 = -\$200
(no poverty gap as income is above poverty line) Poverty Gap Ratio = 0

Problem 2: Headcount Ratio

Question: In Country Y, the poverty line is set at $10,000 per year. A survey of 100 households reveals the following incomes (in dollars per year):

- 40 households earn less than $10,000 per year.
- 60 households earn more than $10,000 per year.

Calculate the **headcount ratio** of poverty.

Problem 2: Headcount Ratio

Solution: The headcount ratio is calculated as the proportion of people living below the poverty line:

$$\text{Headcount Ratio} = \frac{\text{Number of Poor Households}}{\text{Total Number of Households}}$$

Here, the number of poor households = 40, total households = 100.

$$\text{Headcount Ratio} = \frac{40}{100} = 0.40$$

Thus, the headcount ratio is 0.40 or 40%.

Problem 3: Poverty Gap Index

Question: Consider a country with the following data:

- The poverty line is \$5,000 per year.
- There are 100 households with the following incomes (in dollars per year):
  - 20 households earn \$2,000 each.
  - 30 households earn \$3,000 each.
  - 50 households earn \$6,000 each.

Calculate the **poverty gap index** for this country.

Problem 3: Poverty Gap Index

Solution: The poverty gap for each household group is calculated as the difference between the poverty line and the income of the households. The poverty gap index is given by:

$$\text{Poverty Gap Index} = \frac{\sum(\text{Poverty Line} - \text{Income}) \times \text{Number of Poor}}{\text{Total Income of All Households}}$$

For the 20 households earning \$2,000:

$$\text{Poverty Gap} = 5000 - 2000 = 3000$$

$\text{Total Poverty Gap for these households} = 3000 \times 20 = 60,000$
For the 30 households earning \$3,000:

$$\text{Poverty Gap} = 5000 - 3000 = 2000$$

$\text{Total Poverty Gap for these households} = 2000 \times 30 = 60,000$

Problem 3: Poverty Gap Index

For the 50 households earning \$6,000, there is no poverty gap since income is above the poverty line.

$$\text{Total Poverty Gap} = 60,000 + 60,000 = 120,000$$

Total income of all households $=$
$20 \times 2000 + 30 \times 3000 + 50 \times 6000 = 40,000 + 90,000 + 300,000 = 430,000$
Poverty Gap Index $=$

$$\text{Poverty Gap Index} = \frac{120,000}{430,000} \approx 0.279$$

Thus, the Poverty Gap Index is approximately 0.279.